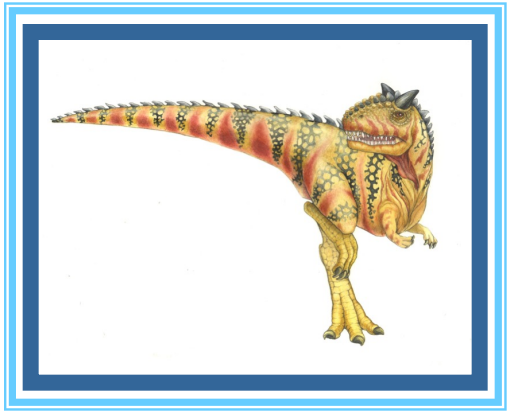
Chapter 3: Inter Process Communication (IPC)



A Unix Shell!

- Loop
  - Read command line from stdin
  - Expand wildcards
  - Interpret redirections < > |
  - pipe (as necessary), fork, dup, exec, wait

Create Your Own Unix Shell

- Command is entered and if length is non-null, keep it in history.
- Parsing : Parsing is the breaking up of commands into individual words and strings
- Checking for special characters like pipes, etc is done
- Checking if built-in commands are asked for.
- If pipes are present, handling pipes.
- Executing system commands and libraries by forking a child and calling execvp.
- Printing current directory name and asking for next input.

Question 6 in Project 1

- Main component:
  - A loop
  - Parsing the user input
  - fork(), execvp()

- Code structure (pseudocode):

```
while (command != “exit”) {
        cout<<“MiniShell>”;
        cin>>command;
        Parsing command;
        pid = fork();
        if (pid < 0) erro;
        else if (pid == 0) execvp();
        else wait();
}
```

Tips for Question 6 of Project 1

- **You need a loop to make a shell always waiting to user input:**

- while(strcmp(command,"exit")!=0){
- cout<<“MiniShell>";
- cin.getline(command,50);
- }

- ***A typical bug:***
  - *Sometimes, if the results do not pop up immediately, but do show up when user enter the next command, then you can try to use fflush(stdout)*

Test “fflush”

```
#include <stdlib.h>
#include <stdio.h>
#include <fcntl.h>
int main(int argc, char **argv)
{
        int pid, status;
        int newfd;      /* new file descriptor */
        if (argc != 2) {
                fprintf(stderr, "usage: %s output_file\n", argv[0]);
                exit(1);
        }
        if ((newfd = open(argv[1], O_CREATIO_TRUNCIO_WRONLY, 0644)) < 0) {
                perror(argv[1]);                /* open failed */
                exit(1);
        }
        printf("This goes to the standard output.\n");
        printf("Now the standard output will go to \"%s\".\n", argv[1]);
        fflush(stdout);
        dup2(newfd, 1);
        printf("This goes to the standard output too.\n");
        exit(0);
}
```

Tips for Question 6 of Project 1

- **How to use execvp:**
- Syntax:
  - int execvp (const char *file, char *const argv[]);
  - file: points to the file name associated with the file being executed.
  - argv: is a null terminated array of character pointers.
- Example
  - char *cmd = "ls";
  - char *argv[3];
  - argv[0] = "ls";
  - argv[1] = "-la";
  - argv[2] = NULL;
  - execvp(cmd, argv);
  - //This will run "ls -la" as if it were a command

Tips for Question 7&8 of Project 1

- **Parsing command:**

```
p=strtok(command, " ");
        while(p&&strcmp(p,"l")!=0){
            command_arrayA[countA]=p;
            countA++;
            p=strtok(NULL," ");
        }
```

Question 8 in Project 1

- How does “ls l wc” work?
- Two steps involved:
  - “ls” is executed, and some intermediate results will be obtained
  - The above intermediate results will be passed by a pipe “l” to become the new input to be processed by “wc”
- How to implement “l” in your own shell?
  - dup2(fd1, fd2) function is needed

Redirecting Standard I/O

A process that communicates solely with another process doesn't use its standard I/O.

- If process communicates with another process only via pipes, redirect standard I/O to the pipe ends
- Functions: ***dup2***

How to use?

- #include <unistd.h>
- int dup2(int *FD1,int FD2*);

Dup2

- For convenience…

```
dup2( int fd1, int fd2 );
```

use `fd2` (`new`) to duplicate `fd1` (`old`)

closes `fd2` if it was in use

- Example: redirect stdin to "foo"

```
fd = open("foo", O_RDONLY, 0);
dup2(fd,0);
close(fd);
```

Dup2 Redirect STDOUT to a File

```
#include <stdio.h>
#include <unistd.h>
#include <fcntl.h>

int main() {
    int file = open("myfile.txt", O_WRONLY | O_CREAT | O_TRUNC, 0644);
    if (file < 0) {
        perror("open");
        return 1;
    }
    // redirecting standard output to the file
    if (dup2(file, STDOUT_FILENO) < 0) {
        perror("dup2");
        close(file);
        return 1;
    }
    // will be redirected to the file
    printf("This will print in myfile.txt\n");
    close(file);
    return 0;
}
```

Redirect STDOUT to File by Executing Command

```
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <fcntl.h>

int main(int argc, char **argv)
{
    int pid, status;
    int newfd;  /* new file descriptor */
    char *cmd[] = { "/bin/ls", "-al", "/", 0 };
    if (argc != 2) {
        fprintf(stderr, "usage: %s output_file\n", argv[0]);
        exit(1);
    }
    if ((newfd = open(argv[1], O_CREATIO_TRUNCIO_WRONLY, 0644)) < 0) {
        perror(argv[1]);    /* open failed */
        exit(1);
    }
    printf("writing output of the command %s to \"%s\"\n", cmd[0], argv[1]);
    dup2(newfd, 1);

    /* now we run the command. It runs in this process and will have */
    /* this process' standard input, output, and error */
    execvp(cmd[0], cmd);
    perror(cmd[0]);     /* execvp failed */
    exit(1);
}
```

Try it to understand how dup2() redirect standard output to a file

An Example of “ls | less”

```
#include <stdio.h>
#include <stdlib.h>
#include <fcntl.h>
#include <unistd.h>
int main(int agrc, char* agrv[])
{
  int pipefds[2];
    pid_t pid;
  if(pipe(pipefds) == -1){
    perror("pipe");
    exit(EXIT_FAILURE);
  }
  pid = fork();
  if(pid == -1){
    perror("fork");
    exit(EXIT_FAILURE);
  }
```

```
if(pid == 0){
  //replace stdout with the write end of the pipe
    dup2(pipefds[1],STDOUT_FILENO);
  //close read to pipe, in child
    close(pipefds[0]);
    execlp("ls","ls",NULL);
    exit(EXIT_SUCCESS);
  }else{
  //Replace stdin with the read end of the pipe
        dup2(pipefds[0],STDIN_FILENO);
  //close write to pipe, in parent
        close(pipefds[1]);
        execlp("less","less",NULL);
        exit(EXIT_SUCCESS);
    }
}
```

Tips for Question 8 of Project 1

- **How to digest "|" in your shell:**

- fork A
  if(pid A<0)
  {...}
  else if(pid A==0)
  { **1. Redirect STDOUT to the write end of the pipe**
- **2. Execute command before "|"** }
  else
  { fork B
  if (pid B<0)
  {...}
  else if (pid B==0)
  {**1. Redirect the read end of the pipe to STDIN**
- **2. Execute the arg after "|"** }
  else
  {...}
  }